

Amin Charepoo

(617) 653 - 0826 | charepoo.a@northeastern.edu | [LinkedIn](#) | [GitHub](#) | [Website](#)

EDUCATION

Northeastern University | Boston, MA

April 2029

Candidate for Bachelor of Computer Engineering

GPA: 3.961 (Dean's List)

Relevant Courses: Embedded Design, Fundamentals of Networks, Discrete Structures, Computing Fundamentals, Differential Equations and Linear Algebra, Calculus 2

Related Activities: Robotics Club, Virtual Reality Lab Assistant, VR Research, Northeastern Virtual Reality Club

TECHNICAL SKILLS

Programming: C++, Python, C#, Java, JavaScript, HTML/CSS

Embedded Systems: ESP32, Arduino, FPGA, GPIO, Hardware Timers, Interrupts, Real-Time Systems, Sensor Integration

Tools: Git, MATLAB, Solidworks, AutoCAD, 3D Printing, Ubuntu Linux, OnShape, Unity3D, VSCode, LaTeX

PROJECTS

Real-Time Cooperative Task Scheduler | *Computing Fundamentals*

Feb 2026 - Apr 2026

- Designed priority-based task scheduler in C++ demonstrating OOP principles.
- Programmed interrupt-driven architecture using GPIO and hardware timer ISRs with safe shared-state management.
- Implemented 5 cooperating classes managing LCD display with VIP preemption and cooperative task resumption.
- Wrote technical documentation in LaTeX covering system design, pseudocode, and implementation analysis.

Autonomous Line-Following Robot | *Northeastern Robotics*

Sept 2025 - Dec 2025

- Programmed an autonomous robot in Arduino C++, processing data from 3 IR sensors and a servo-mounted ultrasonic sensor to drive 4 independently controlled wheel motors for path-following and obstacle detection.
- Developed modular and maintainable code with real-time logic for reliable hardware-software integration.
- Used Git for version control and documented system design to support collaboration and future development.
- Won **1st** place in end-of-year competition against 3 teams as a member of a 2-person team.

Spotify Controller Hub | *Personal Project*

Sept 2025 - Nov 2025

- Designed and built an embedded controller that manages hardware inputs to control Spotify playback.
- Programmed an ESP32 in C++ to handle Wi-Fi communication, API requests, and hardware input processing.
- Created and tested circuit connections for reliable input handling and Wi-Fi connectivity.
- Improved workflow efficiency by enabling seamless hands-free Spotify control, reducing the need to switch applications.

EXPERIENCE

Northeastern University Robotics | *Intro to Robotics Teacher*

Boston, MA | Jan 2026 - Present

- Teach introductory robotics concepts to **50** Northeastern students, including circuits, CAD, Arduino, sensors, and C++, through hands-on projects and demonstrations.
- Guide students through debugging and troubleshooting hardware and software issues in robotics projects.

Northeastern First-Year Engineering Makerspace | *Red Vest*

Boston, MA | Sept 2026 - Present

- Assist first-year engineering students with CAD, programming, circuit wiring, and woodworking projects.
- Maintain and organize makerspace supplies and tools to ensure smooth operation for student use.

Northeastern Virtual Reality | *Executive Board Member, Project Lead*

Boston, MA | Jan 2024 - June 2026

- Led a team of **5** students in developing a VR dungeon-fighting game using Unity3D, C#, and an Oculus Quest 2.
- Coordinated project development, assigned tasks, and managed collaborative workflows through Git.

NEU Virtual Reality Immersive Media Lab | *Work Study Associate*

Boston, MA | May 2025 – June 2025

- Collaborated with a team of **3** to develop a virtual reality story experience in Unity3D using C#.
- Assisted students and faculty with equipment setup and troubleshooting.
- Managed version control and collaborative workflows through Git.

TechHunt PCs | *Business Owner*

Boston, MA | Sept 2021 - Jan 2025

- Founded and operated a small business specializing in building, repairing, and selling custom PCs.
- Built and refurbished over **60** computer systems, ensuring reliable hardware operation and performance.
- Diagnosed and resolved hardware and software issues across diverse PC systems, earning **20+** five-star reviews.

ACTIVITIES AND LEADERSHIP

- Boston Ruhi Institute | Boston Junior Youth Spiritual Empowerment Program Coordinator

2018 – Present